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Architectural fixture connection system

Abstract

An architectural fixture connection system offers acoustical and aesthetic benefits. This system utilizes modular construction, allowing interconnection of a variety of different beams to form an installation that may attach to a wall or ceiling of a building. This system may be formed of a series of sheets which are formed into beams using securing brackets and attachment brackets, then coupled to one another using a series of joiner brackets. Interconnection between adjacent beams need not be linear, allowing free-form shapes and angles.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) The present application claims the benefit of U.S. Provisional Patent Application Ser. No. 63/278,583, filed Nov. 12, 2021, the entireties of which are incorporated herein by reference.

FIELD OF THE INVENTION

(1) The present invention relates generally to the field of ceiling and wall fixtures. More particularly, the present invention relates to an architectural fixture connection system.

BACKGROUND

(2) Fixtures have conventionally provided only horizontally oriented surfaces or vertically oriented planar segments.

(3) Co-owned U.S. Pat. No. 8,733,053 discloses systems and methods for supported architectural designs. Co-owned U.S. Pat. No. 8,782,987 discloses supported architectural structures.

(4) There is a need for new types of acoustical ceiling and wall architectural fixtures. There is a further need for an improved architectural fixture providing sound-absorption. There is an additional need for an improved architectural fixture that provides a modular construction. There is also a need for an improved architectural fixture that provides improved connection between components of the fixture. There is a need for an improved architectural fixture that provides for various configurations of the fixture. There is a further need for an improved architectural fixture that provides routing and/or support for various objects. There is an additional need for a holder that is easier to manufacture, assemble, adjust, and maintain. The present invention satisfies these needs and provides other related advantages.

SUMMARY

(5) An architectural fixture described herein provides sound-absorption. An architectural fixture

described herein provides a modular construction. An architectural fixture described herein provides improved connection between components of the fixture. An architectural fixture described herein provides for various configurations of the fixture. An architectural fixture described herein provides easier manufacture, assembly, adjustment, and maintenance.

(6) In one implementation, the invention is an architectural fixture assembly having a beam extending along a longitudinal axis. The beam has a bottom side, a first sidewall extending from the bottom side to a distal end, and a second sidewall that extends from the bottom side to a distal end. A cavity is formed by inner surfaces of the bottom side, the first sidewall, and the second sidewall. A first elongated channel is formed into the inner surface of the first sidewall. A second elongated channel is formed into the inner surface of the second sidewall. A third fastener extends through the first elongated channel. A second fastener extends through the second elongated channel. A first securing element extends from the first elongated channel to the second elongated channel. The first securing element engages the first fastener and the second fastener.

(7) In another implementation, the invention is an architectural fixture assembly having a first beam extending along a longitudinal axis and a second beam extending along the longitudinal axis. The first beam has a bottom side, a first sidewall extending from the bottom side to a distal end, and a second sidewall that extends from the bottom side to a distal end. A cavity is formed by inner surfaces of the bottom side, the first sidewall, and the second sidewall. A first notch is formed into the inner surface of the first sidewall at a first end of the first sidewall. The second beam has a bottom side, a first sidewall extending from the bottom side to a distal end, and a second sidewall that extends from the bottom side to a distal end. A cavity is formed by inner surfaces of the bottom side, the first sidewall, and the second sidewall. A second notch is formed into the inner surface of the first sidewall at a second end of the first sidewall. A first joiner bracket engages the first notch of the first beam and the second notch of the second beam, the joiner bracket securing the first end of the first sidewall of the first beam to the second end of the first sidewall of the second beam.

(8) In yet another implementation, the invention is an architectural fixture assembly having a first beam extending along a first longitudinal axis and a second beam extending along a second longitudinal axis. The first beam has a bottom side, a first sidewall extending from the bottom side to a distal end, and a second sidewall that extends from the bottom side to a distal end. A cavity is formed by inner surfaces of the bottom side, the first sidewall, and the second sidewall. The second beam has a bottom side, a first sidewall extending from the bottom side to a distal end, and a second sidewall that extends from the bottom side to a distal end. A cavity is formed by inner surfaces of the bottom side, the first sidewall, and the second sidewall. The bottom side, the first sidewall, and the second sidewall of both of the first and second beams are formed from a monolithic, unitary, sheet.

(9) In still another implementation, the invention is an architectural fixture assembly having a first beam extending along a first longitudinal axis and a second beam extending along a second longitudinal axis, the second longitudinal axis oriented at an angle to the first longitudinal axis. The first beam has a bottom side, a first sidewall extending from the bottom side to a distal end, and a second sidewall that extends from the bottom side to a distal end. A cavity is formed by inner surfaces of the bottom side, the first sidewall, and the second sidewall. First and second elongated slots are formed into the first sidewall, the first and second elongated slots extend from the distal end toward the bottom side. The second beam has a bottom side, a first sidewall extending from the bottom side to a distal end, and a second sidewall that extends from the bottom side to a distal end. A cavity is formed by inner surfaces of the bottom side, the first sidewall, and the second sidewall. A first elongated slot is formed into the first sidewall and extends upward from the bottom side. A second elongated slot is formed into the second sidewall and extends upward from the bottom side. The second beam is assembled to the first beam such that the first and second elongated slots of the first beam engage the first and second sidewalls of the second beam.

(10) In another implementation, the invention is an architectural fixture assembly having a first

beam extending along a first longitudinal axis, a second beam extending along a second longitudinal axis, the second longitudinal axis oriented at a first angle to the first longitudinal axis, and a third beam extending at a third longitudinal axis, the third longitudinal axis at a second angle to the first longitudinal axis. The first beam has a bottom side, a first sidewall extending from the bottom side to a distal end, and a second sidewall that extends from the bottom side to a distal end. A cavity is formed by inner surfaces of the bottom side, the first sidewall, and the second sidewall. First, second, and third elongated slots are formed into the first sidewall, the first, second, and third elongated slots extending from the distal end toward the bottom side. The second beam has a bottom side, a first sidewall extending from the bottom side to a distal end, and a second sidewall that extends from the bottom side to a distal end. A cavity is formed by inner surfaces of the bottom side, the first sidewall, and the second sidewall. A first elongated slot is formed into the first sidewall and extends upward from the bottom side. A second elongated slot is formed into the second sidewall and extends upward from the bottom side. The third beam has a bottom side, a first sidewall extending from the bottom side to a distal end, and a second sidewall that extends from the bottom side to a distal end. A cavity is formed by inner surfaces of the bottom side, the first sidewall, and the second sidewall. A first elongated slot is formed into the first sidewall and extends upward from the bottom side. A second elongated slot is formed into the first sidewall and extends downward from the distal end. A third elongated slot is formed into the second sidewall and extends downward from the distal end. The second beam is assembled to the first beam such that 1) the first elongated slot of the first beam engages the first sidewall of the second beam, 2) the second elongated slot of the first beam engages the second sidewall of the second beam and the first sidewall of the third beam, and 3) the third elongated slot of the first beam engages the second sidewall of the third beam.

(11) In yet another implementation, the invention is an architectural fixture assembly having a first beam extending along a first longitudinal axis and a second beam extending along a second longitudinal axis. The first beam has a bottom side extending along a first plane, a first sidewall extending from the bottom side to a distal end, and a second sidewall that extends from the bottom side to a distal end. A cavity is formed by inner surfaces of the bottom side, the first sidewall, and the second sidewall. The second beam has a bottom side extending along a second plane, a first sidewall extending from the bottom side to a distal end, and a second sidewall that extends from the bottom side to a distal end. A cavity is formed by inner surfaces of the bottom side, the first sidewall, and the second sidewall. The second plane is oriented at an acute angle with respect to the first plane.

(12) In a further implementation, the invention is an architectural fixture assembly having a first beam extending along a first longitudinal axis, a second beam extending along a second longitudinal axis, and a third beam extending at a third longitudinal axis. The first beam has a bottom side, a first sidewall extending from the bottom side to a distal end, and a second sidewall that extends from the bottom side to a distal end. A cavity is formed by inner surfaces of the bottom side, the first sidewall, and the second sidewall. The second beam has a bottom side, a first sidewall extending from the bottom side to a distal end, and a second sidewall that extends from the bottom side to a distal end. A cavity is formed by inner surfaces of the bottom side, the first sidewall, and the second sidewall. The third beam has a bottom side, a first sidewall extending from the bottom side to a distal end, and a second sidewall that extends from the bottom side to a distal end. A cavity is formed by inner surfaces of the bottom side, the first sidewall, and the second sidewall. A first joiner bracket engages the first sidewall of the first beam and the second sidewall of the second beam. A second joiner bracket engages the first sidewall of the second beam and the second sidewall of the third beam.

(13) Further areas of applicability of the present invention will become apparent from the detailed description provided hereinafter. It should be understood that the detailed description and specific

examples, while indicating the preferred embodiment of the invention, are intended for purposes of illustration only and are not intended to limit the scope of the invention.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The various present embodiments now will be discussed in detail with an emphasis on highlighting the advantageous features with reference to the drawings of various embodiments. The illustrated embodiments are intended to illustrate, but not to limit the invention. The drawings are not to be construed as drawn to scale.
- (2) FIG. 1 is a top, front, right perspective view of an architectural fixture embodying the invention.
- (3) FIG. 2 is a front side elevation view of the beam of FIG. 1.
- (4) FIG. 3 is a top plan view of the beam of FIG. 1.
- (5) FIG. 4 is a perspective view of an attachment bracket as may be utilized in the beam of FIG. 1.
- (6) FIG. 5 is a top, front, right perspective view of an additional architectural fixture embodying the invention.
- (7) FIG. 6 is a front side elevation view of the beam of FIG. 5.
- (8) FIG. 7 is a cross-sectional view of the beam of FIG. 5.
- (9) FIG. 8 is a top plan view of the beam of FIG. 5.
- (10) FIG. 8A illustrates a manner of attaching the architectural fixture of FIG. 5 to a support structure.
- (11) FIG. 8B illustrates a top view of a securing element as may be used in the beam of FIGS. 1 and 5.
- (12) FIG. 9 is a top, front, left perspective view of a connection key with and end cover embodying the invention.
- (13) FIG. 10 is a front side elevation view of the connection key with and end cover of FIG. 9.
- (14) FIG. 11 is a left side elevation view of the connection key with and end cover of FIG. 9.
- (15) FIG. 12 is a top plan view of the connection key with and end cover of FIG. 9.
- (16) FIG. 13 is a top, front, right perspective view of an additional architectural fixture shown in engagement with a second beam embodying the invention.
- (17) FIG. 13A illustrates the architectural fixture of FIG. 13 whereby the two beams are coupled together by joiner brackets.
- (18) FIG. 14 is a front side elevation view of an architectural fixture shown in engagement with a second beam of FIG. 13.
- (19) FIG. 15 is a top plan view of an architectural fixture shown in engagement with a second beam of FIG. 13.
- (20) FIG. 16A is a perspective view of two joiner brackets as may be used in the architectural fixture of FIG. 13.
- (21) FIG. 16B is a right side view of the two joiner brackets of FIG. 16A.
- (22) FIG. 16C is a front side view of the two joiner brackets of FIG. 16A.
- (23) FIG. 16D is a top plan view of the two joiner brackets of FIG. 16A.
- (24) FIG. 17 is a top, front, right perspective view of an additional architectural fixture embodying the invention wherein two beams meet at an angle.
- (25) FIG. 17A is a top plan view of the beams of FIG. 17 in an unfolded state.
- (26) FIG. 17B is perspective view of the beams of FIG. 17 in a partially folded state.
- (27) FIG. 18 is a front side elevation view of the beams of FIG. 17.
- (28) FIG. 19 is a top plan view of the beams of FIG. 17.
- (29) FIG. 20 is a perspective view of the beams of FIG. 17 showing the meeting of the two beams.

- (30) FIG. 21 is a detail view showing a securing element for assembling the two beams.
- (31) FIG. 21B is a top perspective view of the beams of FIG. 17.
- (32) FIG. 21C is an illustration showing different angled configurations for the architectural fixture of FIG. 17.
- (33) FIG. 21D is a side view showing the securing element of the architectural fixture of FIG. 17.
- (34) FIG. 22 is a top, front, right perspective view of an additional architectural fixture embodying the invention wherein the beams are configured to form a triangle-like shape.
- (35) FIG. 23 is a front side elevation view of the beams of FIG. 22.
- (36) FIG. 24 is a top plan view of the beams of FIG. 22.
- (37) FIG. 25 is a top, front, right perspective view of an additional architectural fixture embodying the invention wherein the beams are configured to form a hexagon-like shape.
- (38) FIG. 26 is a front side elevation view of the beams of FIG. 25.
- (39) FIG. 27 is a top plan view of the beams of FIG. 25.
- (40) FIG. 28 is an exploded top, front, right perspective view of an additional architectural fixture embodying the invention wherein two beams are shown in engagement.
- (41) FIG. 29 is a front side elevation view of the beams of FIG. 28.
- (42) FIG. 30 is a top plan view of the beams of FIG. 28.
- (43) FIG. 31 is a top, front, right perspective view of an additional architectural fixture embodying the invention wherein three beams are shown in engagement.
- (44) FIG. 31A is an exploded view of the architectural fixture of FIG. 31.
- (45) FIG. 31B is a second exploded view of the architectural fixture of FIG. 31.
- (46) FIG. 32 is a front side elevation view of the beams of FIG. 31.
- (47) FIG. 33 is a top plan view of the beams of FIG. 31.
- (48) FIG. 33A is a bottom plan view of the architectural fixture of FIG. 31.
- (49) FIG. 34 is a top, front, right perspective view of an additional architectural fixture embodying the invention wherein six beams are shown in engagement.
- (50) FIG. 35 is a front side elevation view of the beams of FIG. 34.
- (51) FIG. 36 is a top plan view of the beams of FIG. 34.
- (52) FIG. 37 is a top, front, right perspective view of an additional architectural fixture embodying the invention.
- (53) FIG. 38 is a front side elevation view of the beams of FIG. 37.
- (54) FIG. 39 is a top plan view of the beams of FIG. 37.
- (55) FIG. 40 is a top, front, right perspective view of an additional architectural fixture embodying the invention.
- (56) FIG. 41 is an enlarged section of FIG. 40 taken along line 41 of FIG. 40.
- (57) FIG. 42 is a top, front, right perspective view of an additional architectural fixture embodying the invention wherein the beams are configured to form a lattice.
- (58) FIG. 43 is a front side elevation view of the beams of FIG. 42.
- (59) FIG. 44 is a top plan view of the beams of FIG. 42.
- (60) FIG. 45 is a top, front, right perspective view of the beams of FIG. 42 shown with dashed lines indicating movement for the beams to engage/disengage.
- (61) FIG. 46 is a top, front, right perspective view of an additional architectural fixture embodying the invention where in the beams are configured to form a star-like shape.
- (62) FIGS. 46A and 46B are partial views of the architectural fixture of FIG. 46.
- (63) FIG. 47 is a front side elevation view of the beams of FIG. 46.
- (64) FIG. 48 is a top plan view of the beams of FIG. 46.
- (65) FIG. 49 is a top, front, right perspective view of an additional architectural fixture embodying the invention wherein the beams are disposed to form a continuous up and down three-dimensional movement configuration.
- (66) FIG. 50 is a front side elevation view of the beams of FIG. 49.

(67) FIG. 51 is a top plan view of the beams of FIG. 49.

(68) FIG. 51A is a bottom perspective view of a portion of the architectural fixture of FIG. 49.

(69) FIG. 51B is a view of one of the beams of the architectural fixture of FIG. 49 in an unfolded configuration.

(70) FIG. 51C is a three-dimensional architectural fixture system formed by connecting a plurality of the architectural fixtures of FIG. 49.

(71) FIGS. 52A and 52B illustrate a top, front, left perspective view of a beam of an architectural fixture embodying the invention, with a beam, and an outwardly facing side of a connection key with an end cover flush with an end of the beam.

(72) FIG. 53 is a front side elevation view of the beam of FIG. 52B.

(73) FIG. 54 is a top plan view of the beam of FIG. 52B.

DETAILED DESCRIPTION

(74) The following description of the preferred embodiment(s) is merely exemplary in nature and is in no way intended to limit the invention, its application, or uses.

(75) The description of illustrative embodiments according to principles of the present invention is intended to be read in connection with the accompanying drawings, which are to be considered part of the entire written description. In the description of embodiments of the invention disclosed herein, any reference to direction or orientation is merely intended for convenience of description and is not intended in any way to limit the scope of the present invention. Relative terms such as “lower,” “upper,” “horizontal,” “vertical,” “above,” “below,” “up,” “down,” “top” and “bottom” as well as derivatives thereof (e.g., “horizontally,” “downwardly,” “upwardly,” etc.) should be construed to refer to the orientation as then described or as shown in the drawing under discussion. These relative terms are for convenience of description only and do not require that the apparatus be constructed or operated in a particular orientation unless explicitly indicated as such. Terms such as “attached,” “affixed,” “connected,” “coupled,” “interconnected,” and similar refer to a relationship wherein structures are secured or attached to one another either directly or indirectly through intervening structures, as well as both movable or rigid attachments or relationships, unless expressly described otherwise. Moreover, the features and benefits of the invention are illustrated by reference to the exemplified embodiments. Accordingly, the invention expressly should not be limited to such exemplary embodiments illustrating some possible non-limiting combination of features that may exist alone or in other combinations of features; the scope of the invention being defined by the claims appended hereto.

(76) Co-owned U.S. patent application Ser. No. 16/863,949, filed Apr. 30, 2020, and published as US2021/0340761 on Nov. 4, 2021, discloses an architectural fixture connection system. The present application is directed, at least in part, to updates, additions, and/or modifications to the architectural fixture connection system described in U.S. patent application Ser. No. 16/863,949. US2021/0340761 is incorporated herein by reference.

(77) Referring first to FIGS. 1-12, an architectural fixture assembly **100** is illustrated in accordance with an embodiment of the present invention. The architectural fixture assembly **100** is a generally U-shaped beam **102** that is formed by folding a flat elongated sheet of material. Optionally, a plurality of V-cuts, grooves, or other features may be formed to facilitate bending of the sheet. Once folded, the U-shaped beam **102** comprises a bottom side **104**, a first sidewall **106** that extends from the bottom side **104** to a distal end **117**, and a second sidewall **108** that extends from the bottom side **104** to a distal end **118**. The bottom side **104** and the first and second sidewalls **106**, **108** collectively define a cavity **105**. A longitudinal axis A-A extends along the beam **102**, extending parallel to the bottom side **104** and the first and second sidewalls **106**, **108**. The beam **102** extends from a first end **103a** to a second end **103b** along the longitudinal axis A-A.

(78) The cavity **105** may be open or closed at opposing ends. That is, the first sidewall **106** extends from a first end **107a** to a second end **107b** and the second sidewall **108** extends from a first end **109a** to a second end **109b**. The cavity **105** is bounded by connection keys **144** at each of the first

and second ends **107a**, **107b**, **109a**, **109b** of the first and second sidewalls **106**, **108**. Optionally, the cavity **105** may be open at one or both of the first and second ends **107a**, **107b**, **109a**, **109b** of the first and second sidewalls **106**, **108**. The use of connection keys **144** is optional as will be described in greater detail below.

(79) The first sidewall **106** has an inner surface **110** that faces the cavity **105** and the second sidewall **108** has an inner surface **112** that faces the cavity **105**. The first sidewall **106** has a first notch **113** formed into its inner surface **110** located adjacent to the first end **107a** and a second notch **114** formed into its inner surface **110** located adjacent to the second end **107b**. The second sidewall **108** has a first notch **115** formed into its inner surface **112** located adjacent to the first end **109a** and a second notch **116** formed into its inner surface **112** located adjacent to the second end **109b**.

(80) Referring to FIGS. **1** and **9-12**, the architectural fixture **100** comprises a pair of connection keys **144** that are coupled to the U-shaped beam **102** to close the openings at the first and second opposing ends of the first and second sidewalls **106**, **108**. Each of the connection keys **144** comprises a pair of protuberances **146** that nest within the notches **113-116** in the inner surfaces **110**, **112** of the first and second sidewalls **106**, **108**. Thus, the connection keys **144** can be positioned with the protuberances **146** aligned with the notches **113-116** and then the connection keys **114** can be slid relative to the beam **102** to couple the connection keys **144** to the beam **102**. The connection key **144** may comprise a plate portion **147** and a connection portion **148**, or may be formed as a single, integrally formed, monolithic, unitary, component. Optionally, the connection keys **144** may bound the cavity **105**.

(81) The architectural fixture **100** is elongated along the longitudinal axis A-A. Furthermore, there is an elongated channel **170** formed into each of the inner surfaces **110**, **112** of the first and second sidewalls **106**, **108**. The elongated channel **170** is only visible in the inner surface **112** of the second sidewall **108** in the figures provided, but it should be appreciated that an identical elongated channel exists in the inner surface **110** of the first sidewall **106**. The elongated channels **170** are elongated in the direction of the longitudinal axis A-A. In other implementations, the elongated channels **170** may be formed as two separate elongated channels **170** formed in the inner surface **110** of the first sidewall **106** and two separate elongated channels **170** formed in the inner surface **112** of the second sidewall **108**. Each of these individual elongated channels **170** may be formed immediately adjacent the connection keys **144** at the first and second ends **107a**, **107b**, **109a**, **109b** of the first and second sidewalls **106**, **108**. In yet other implementations, there may be more than two elongated channels **170** on each of the first and second sidewall.

(82) In the exemplified embodiment, the elongated channels **170** are spaced inwardly of the first and second ends **107a**, **107b**, **109a**, **109b** of the first and second sidewalls **106**, **108**, but the elongated channels **170** could extend to the various ends of the first and second sidewalls **106**, **108** in other embodiments. More specifically, in the exemplified embodiment when the connection keys **144** are coupled to the beam **102**, the ends of the elongated channels **170** are spaced from the connection keys **144**. The elongated channels **170** do not extend through the full thickness of the first and second sidewalls **106**, **108** in the exemplified embodiment, although they may in other embodiments.

(83) The architectural fixture **100** further comprises a pair of attachment brackets **180A**, **180B**. FIGS. **1-3** illustrate the architectural fixture **100** with the attachment brackets **180A** and FIGS. **5-8A** illustrate the architectural fixture **100** with the attachment brackets **180B**. The architectural fixture **100** is configured to receive either of the different types of attachment brackets **180A**, **180B**. Referring to FIGS. **1-3**, the attachment brackets **180A** are a ceiling hanging bracket that comprises arm portions **181A** that nest within the elongated channels **170** in the first and second sidewalls **106**, **108** and a central portion **182A** that extends between the arm portions **181A**. The body portion **182A** comprises an aperture or through-hole within which a connection feature of a rod or cable **101** may be attached to the attachment bracket **180A**. The rod or cable **101** is affixed to the ceiling

(either directly or indirectly) and then is attached to the attachment bracket **180A**, which is in turn attached to the architectural fixture **100** by nesting the arm portions **181A** of the attachment bracket **180A** within the elongated channels **170** of the beam **102**. The attachment brackets **180A** may be configured to slide in the direction of the longitudinal axis A-A while remaining coupled to the beam **102**.

(84) FIG. **4** illustrates one of the attachment brackets **180A**. As can be seen, the attachment bracket **180A** has a central portion **182A** and arm portions **181A** extending upwardly from the central portion **182A**. The central portion **182A** incorporates an aperture or hole which receives a cable, rod, or other attachment device. The cable, rod, or other attachment device allows suspending the beam **102** from a roof or ceiling, the beam **102** being spaced from the roof or ceiling. When the beam **102** is mounted, the central portion **182** is approximately horizontal, while the arm portions **181A** engage the elongated channels **170**. The upward extension of the arm portions **181A** allow the attachment brackets **180A** to be snapped into position and engage the elongated channels **170** without the need for disassembly of the beam **102**.

(85) Still referring to FIGS. **1-3**, there is also a pair of securing elements **190** (i.e., a hook bracket) mounted to the U-shaped beam **102** within the elongated channels **170**. FIG. **8B** also illustrates one of the securing element **190** in greater detail. The securing elements **190** are coupled to the U-shaped beam **102** along opposing ends of the elongated channels **170** in the exemplified embodiment. The securing elements **190** have a main body **192** and a pair of hooks **191**, each hook **191** extending from opposite ends of the main body **192**. The hooks **191** nest within the elongated channels **170** and hold the U-shaped beam **102** in its folded U-shaped configuration. Thus, the U-shaped beam **102** may be capable of maintaining its U-shape without the use of glue or other fasteners, simply with the use of the securing elements **190**.

(86) The hooks **191** engage fasteners **171** which are fitted into holes **172** formed in the first and second sidewalls **106**, **108**. Optionally, the fasteners **171** may be pins, screws, bolts, or any other device configured to receive the hooks **191** of the securing element. The holes **172** intersect the elongated channels **170** and the fasteners extend through the elongated channels **170**. The securing elements **190** extend transverse to the longitudinal axis A-A. Optionally, the attachment brackets **180A** may engage different elongated channels **170** from the elongated channels **170** engaged by the securing elements **190**.

(87) Referring to FIGS. **5-8A**, the architectural fixture **100** is illustrated with the second type of attachment bracket **180B**. FIGS. **5-8A** are identical to FIGS. **1-3** except that the attachment brackets **180B** are used instead of the attachment brackets **180A**. The attachment brackets **180B** are configured for direct mounting. Thus, the attachment brackets **180B** comprise arm portions **181B** that nest within the elongated channels **170** and a central portion **182B** that extends between the arm portions **181B**. The arm portions **181B** comprise hooks which aid in engagement with the elongated channels **170**. There are apertures in the central portion **182B** through which fasteners such as screws, bolts, or the like may extend for purposes of directly coupling the attachment bracket **180B** to a support structure (a ceiling, wall, T-grid, Unistrut, threaded-rod cable, or the like). Optionally, the attachment brackets **180B** may engage different elongated channels **170** from the elongated channels **170** engaged by the securing elements **190**.

(88) FIG. **8A** illustrates the architectural fixture **100** of FIGS. **5-8A** being attached to a support structure in greater detail. Specifically, first the attachment brackets **180B** are attached to the support structure. The support structure may be a horizontal surface such as a ceiling or a vertical surface such as a wall. Next, the U-shaped beam **102** is slidably attached to the attachment brackets **180B** while the arm portions **181B** of the attachment brackets **180B** nest within the elongated channels **170**. Finally, the connection keys **144** are attached to the ends of the U-shaped beam **102**. In other implementations, the connection keys **144** are attached prior to coupling of the beam **102** to the attachment brackets **180B**.

(89) Referring to FIGS. **13-15**, another embodiment of an architectural fixture **200** is illustrated.

The architectural fixture **200** comprises two U-shaped beams **102a**, **102b** coupled together in a side-by-side manner. The two U-shaped beams **102a**, **102b** are identical to the U-shaped beams of FIGS. **1-12**, utilizing the attachment bracket **180A** of FIGS. **1-3**. In this embodiment, the two U-shaped beams **102a**, **102b** are attached together with a pair of joiner brackets **130**. In particular, the first U-shaped beam **102a** comprises a first end **103a** and the second U-shaped beam **102b** comprises a second end **103b**. In this variation, the connection keys **144** are not positioned at the first end **103a** of the first U-shaped beam **102a** or at the second end **103b** of the second U-shaped beam **102b**. Thus, the cavities **105** of each of the beams **102a**, **102b** are only bounded on one end of each of the beams **102a**, **102b**.

(90) As illustrated in FIGS. **16A-D**, the joiner brackets **130** each comprise a body portion **131**, a first flange **132** located along a first edge of the body portion **131**, and a second flange **133** located along a second edge of the body portion **131**. The first and second flanges **132**, **133** are elongated in a vertical direction. The joiner brackets **130** are coupled to the two U-shaped beams **102a**, **102b** within the notches **114a**, **116a**, **113b**, **115b** that were previously described as being used for coupling the connection keys **144**. That is, a first one of the joiner brackets **130** is positioned so that the first flange **132** nests within the notch **114a** of the first U-shaped beam **102a** and the second flange **133** nests within the notch **113b** of the second U-shaped beam **102b**. Similarly, a second one of the joiner brackets **130** is positioned so that the first flange **132** nests within the notch **116a** of the first U-shaped beam **102a** and the second flange **133** nests within the notch **115b** of the second U-shaped beam **102b**. In this way, the joiner brackets **130** hold the first and second U-shaped beams **102a**, **102b** into the adjacent side-to-side position due to their engagement with the various notches as described herein. The joiner brackets **130** apply pressure onto the notches **113-116** to hold the two U-shaped beams **102** together. As shown in FIG. **13A**, there may be four of the joiner brackets **130** used to attach to the two U-shaped beams **102a**, **102b** in some embodiments. Alternatively, the joiner brackets **130** may be made longer to achieve a more secure attachment between the two U-shaped beams. Greater or fewer joiner brackets **130** may be used to couple beams **102** as desired.

(91) Referring to FIGS. **17-21D**, an architectural fixture **300** is illustrated in accordance with another embodiment. The architectural fixture **300** comprises a first beam **301a** and a second beam **301b**. The fixture **300** is alterable from an unfolded configuration as shown in FIG. **17A** to a folded configuration as shown in FIG. **17**. Thus, the beams **301a**, **301b** of the architectural fixture **300** are not formed by coupling multiple components together, but are instead a monolithic structure formed by folding a flat panel. The beams **301a**, **301b** are formed as a single, integrally formed, unitary, monolithic sheet. Thus, the beams **301a**, **301b** are formed of a single piece of material. The architectural fixture **300** is formed as two beams **301a**, **301b** that intersect at an angle. A first longitudinal axis A-A extends along the beam **301a**, while a second longitudinal axis B-B extends along the beam **301b**. The angle between the two beams **301a**, **301b** is an angle defined by the internal corner formed by the intersection of the two beams **301a**, **301b**. The angle may be acute or obtuse, and may be substantially any angle. For instance, the angle may be 60 degrees or less. Alternately, the angle may be 90 degrees or may be 137 degrees or greater. Various exemplary angles are illustrated in FIG. **21C**.

(92) Referring to FIG. **17A**, when unfolded the architectural fixture **300** comprises a bottom wall portion **302** which comprises a first bottom wall **302a** and a second bottom wall **302b** that are separated by a V-shaped recess **350**. A first sidewall **306** extends from a first end of the bottom wall portion **302** and a second sidewall **308** extending from a second end of the bottom wall portion **302**. The first sidewall **306** is divided into two portions **306a**, **306b** by a folding line **360**. The first portion **306a** of the first sidewall **306** is connected to the first bottom wall **302a** and the second portion **306b** of the first sidewall **306b** is connected to the second bottom wall **302b**. The second sidewall **308** comprises a first portion **308a** and a second portion **308b** that are separated from one another.

(93) The first and second portions **308a**, **308b** of the second sidewall **308** comprise tabs **340** on their ends which face each other. FIG. **17B** illustrates the fixture **300** in a partially folded state between the unfolded state of FIG. **17A** to the folded state of FIG. **17**. The first and second portions **306a**, **306b** of the first sidewall **306** are folded about the fold line that exists between the first sidewall **306** and the bottom wall portion **302**. The first and second portions **308a**, **308b** of the second sidewall **308** are folded relative to the bottom wall portion **302**. Next, the first sidewall **306** is folded about the folding line **360** which closes the V-shaped recess and brings the tabs **340** along the ends of the first and second portions **308a**, **308b** of the second sidewall **308** near to each other so that they can be coupled together. Securing elements **390** that are the same as the securing elements **190** can then be used to maintain the architectural fixture **300** in the folded configuration. The architectural fixture includes notches that allow for coupling of connection keys **344** (which are identical to the connection keys **144** described above)

(94) FIGS. **20**, **21**, and **21D** illustrate an additional securing element **330** which may be used with the architectural fixture **300**. The securing element **330** is a plate that slides into a notch **331** between the tabs/flaps **340** to hold or maintain the angle of the architectural fixture **300**. FIG. **21B** is a view of the architectural fixture **300** with the securing element **330** and the securing elements **190** helping to hold the architectural fixture **300** in the folded configuration. FIG. **21C** illustrates different configurations to which the architectural fixture **300** may be folded in different embodiments. FIG. **21D** illustrates the securing element **330**. The securing element **330** has a base portion **333** and two fingers **332**. The base portion **333** is configured for grasping by a user and fits within the notch **331**, while the fingers **332** extend beyond the notch **331** and engage the tabs **340** to prevent separation of the tabs **340**. As can be seen, each of the tabs **340** has two notches **331** but greater or fewer notches **331** may be formed and all notches **331** need not receive a securing element **330**.

(95) Referring to FIGS. **22-24**, an architectural fixture **400** is illustrated in accordance with another embodiment of the present invention. The architectural fixture **400** comprises three separate architectural fixtures **300**, with architectural fixtures **300a-c** attached together to form the architectural fixture **400**. The architectural fixtures **300a-c** are identical to the architectural fixture **300** described above with reference to FIGS. **17-21D**. For the architectural fixture **400**, beams **301a-f** are attached together at their ends with joiner brackets **430**. The joiner brackets **430** are identical to the joiner brackets **130** described above. That is, the beams **301a-f** comprise notches that interact with flanges of the joiner brackets **430** to facilitate the coupling of the beams **301a-f** together. Remaining features of the architectural fixture **400** are the same as features previously described and therefore they are not described herein in the interest of brevity. However, the similar features can be readily identified by viewing and comparing the figures.

(96) Referring to FIGS. **25-27**, an architectural fixture **500** is illustrated in accordance with yet another embodiment of the present invention. The architectural fixture **500** comprises six of the architectural fixtures **300** described above with reference to FIGS. **17-21D**. However, the beams **301** have a different beam angle than the beams **301a**, **301b** described above with reference to FIGS. **17-21D**. As noted and shown in FIG. **21C**, the beams **301a**, **301b** may be bent at various different angles. Depending on the angle of the beams **301**, a specific number of them may be arranged in a loop and coupled together to form an architectural fixture, such as the architectural fixture **500**. Again, joiner brackets **530** (identical to the joiner brackets **130** previously described) and securing elements **590** (identical to the securing elements **190** previously described) may be used in the manner described above to maintain each of the beams **301** in their folded configuration and to facilitate the coupling of the adjacent beams **301** to one another.

(97) Referring to FIGS. **28-30**, an architectural fixture **600** is illustrated in accordance with another embodiment of the present invention. The architectural fixture **600** comprises a first beam **610** and a second beam **620** that are configured to be coupled together in a T-shape. The first and second beams **610**, **620** are both linearly elongated beams in the exemplified embodiment. The first beam

610 extends along a first longitudinal axis A-A while the second beam **620** extends along a second longitudinal axis B-B. The first beam **610** is much like the beam **102** shown in FIG. 1, and both of its ends are closed by a connection key **611**. The first beam **610** comprises a first sidewall **612** and a second sidewall **613** that extend upwardly from a bottom wall portion **616**. The first sidewall **612** terminates in a distal end **614**. Furthermore, the first sidewall **612** comprises two elongated slots **615** that extend downwardly from the distal end **614** of the first sidewall **612** in a direction towards the floor portion. The two elongated slots **615** are spaced apart from one another by a predetermined distance.

(98) The second beam **620** comprises a bottom wall portion **621**, a first sidewall **622**, and a second sidewall **623**. The second beam **620** is closed at one end by an end wall **630**, but remains open at the opposite end. The first sidewall **622** comprises an extension portion **624** that protrudes beyond the bottom wall portion **621** and the second sidewall **623** comprises an extension portion **625** that protrudes beyond the bottom wall portion **621**. The extension portions **624**, **625** are cantilevered from the bottom wall portion **621**. A first elongated slot **626** is formed into the first extension portion **624** adjacent to where the first extension portion **624** meets the bottom wall portion **621**. A second elongated slot **627** is formed into the second extension portion **625** adjacent to where the second extension portion **625** meets the bottom wall portion **621**. Each of the first and second elongated slots **626**, **627** extends upwardly from a lower edge of the first and second extension portions **625**, **626**, respectively. The first and second beams **610**, **620** may be coupled together by mating the elongated slots **614**, **615** of the first beam **610** with the elongated slots **626**, **627** of the second beam **620**, as indicated by the dashed line in FIG. 26. Thus, the first and second beams **610**, **620** may be held together using friction only, or additional fasteners or brackets could be included to create a more secure attachment between the first and second beams **610**, **620**.

(99) Referring to FIGS. 31-33A, an architectural fixture **700** is illustrated in accordance with an embodiment of the present invention. The architectural fixture **700** comprises three beams **710**, **720**, **730**. The beam **710** comprises three elongated slots **711**, **712**, **713** along one of its sidewalls, each of the elongated slots **711**, **712**, **713** extending downwardly from a distal end of the sidewall. The elongated slots **711**, **713** extend through the sidewall of the beam **710** at an oblique angle relative to a longitudinal axis A-A of the beam **710**. The slot **712** is larger in width than the slots **711**, **713** to receive parts of both of the beams **720** and **730**.

(100) The beam **730** extends along a longitudinal axis C-C and has three elongated slots **731**, **732**, **733**. The elongated slot **731** extends from a lower edge of the beam **730** upwardly for less than half of the length of the beam **730**. The slot **732** is aligned with the slot **731**, but extends downwardly from an upper edge of the beam **730** for less than half of the length of the beam **730**. Thus, the slots **731**, **732** are aligned, but spaced apart. The slot **733** extends upwardly from the lower edge of the beam **730** along the other sidewall of the beam **730**. The beam **730** is connected to the beam **710** by slidable mating between the slot **731** of the beam **730** and the slot **712** of the beam **710** and slidable mating between the slot **733** of the beam **730** and the slot **713** of the beam **710**.

(101) The beam **720** extends along a longitudinal axis B-B and comprises two elongated slots **721**, **722**. After the beams **710**, **730** are coupled together, the beam **720** is connected to the beams **710**, **730** by slidable mating between the slot **721** of the beam **720** and the slot **711** of the beam **710** and slidable mating between the slot **722** of the beam **720** and the slot **732** of the beam **730**. The slot **722** of the beam **720** also nests within the slot **712** of the beam **710**.

(102) The various elongated slots may be shaped, sized, and otherwise configured and angled to enable parts of the other beams to be received therein. Thus, for example, the slot **712** of the beam **710** is somewhat larger than the other slots because parts of each of the beams **720**, **730** are received in the slot **712**. Moreover, the edges of the slots may be angled relative to the longitudinal axis of the beam in which the slot is formed to allow the nesting arrangement as shown in FIGS. 31-33A. FIG. 33A illustrates the end result from a bottom plan view, which is the view that would be available to a person inside of a room in which the architectural fixture **700** is hanging.

(103) FIGS. **34-36** illustrate an architectural fixture **800** in accordance with an embodiment of the present invention. The architectural fixture **800** uses much of the same technology as described above for the architectural fixture **700**, except that more beams and slots are used. In particular, in the architectural fixture **800**, there are five beams **810**, **820**, **830**, **840**, **850** each with a variety of slots to enable them to be coupled together similar to the manner described above with reference to the architectural fixture **700**.

(104) One difference in the architectural fixture **800** as compared to the architectural fixture **700** is that the slots in the main beam **810** are oriented perpendicular to the longitudinal axis A-A of the main beam **810**. In the architectural fixture **700**, the slots **711**, **713** in the main beam **710** are oriented at an angle relative to the longitudinal axis A-A of the beam **710**. Thus, in this embodiment each of the beams **820**, **830**, **840**, **850** comprises (numbering provided for the beam **820** only, but it is the same for the other beams) first and second sidewalls **821**, **822**. The first sidewall **821** comprises a main portion **823** and a connection portion **824** that is coupled to the main portion **823** with a living hinge. The second sidewall **822** also comprises a main portion **825** and a connection portion **826** that is coupled to the main portion **825** with a living hinge. Thus, the connection portions **824**, **826** can bend or flex or pivot relative to the main portions **823**, **825**. Furthermore, the connection portions **824**, **826** are the portions that are engaged with the main beam **810** for purposes of coupling the beams **810**, **820** together. This allows the slots in the main beam **810** to be oriented perpendicular to the longitudinal axis of the main beam **810** while still forming the double “K” shape structure as the architectural fixture **800** because the connection portions **824**, **826** can pivot to facilitate slidable mating with the slots of the main beam **810**. Although the invention is described with the main and connection portions attached with a living hinge, in other embodiments there may not be a living hinge, but the connection portions may be bent/angled relative to the main portions to facilitate their insertion into the slots that are oriented perpendicular to the longitudinal axis of the main beam.

(105) Referring now to FIGS. **37-39**, an architectural fixture **900** is illustrated in accordance with an embodiment of the present invention. In this embodiment, there are two beams **910**, **920** that intersect and are attached together via slots (not visible) on each of the beams **910**, **920**. The first beam **910** extends along a longitudinal axis A-A while the second beam **920** extends along a longitudinal axis B-B. In particular, for the beam **910** the slots extend from the bottom side upwards and for the beam **920** the slots extend from the top side downwards. In this embodiment, there are channels **970** on the inner surfaces of the sidewalls of the beams **910**, **920** to receive attachment brackets **980** (like the attachment brackets **180A**, **180B** described above with reference to FIGS. **1-11**). The channels **970** allows for adjustability, and quick, fastenerless connections of the attachment locations for the attachment brackets **980** and the securing elements **990**. Connection keys **944** may be included as well as discussed previously in this disclosure.

(106) Referring to FIGS. **40** and **41**, an architectural fixture **1000** is illustrated in accordance with an embodiment of the present invention. The architectural fixture **1000** includes channels on the inner surfaces of the beams for receiving attachment brackets and securing elements, as has been described above. Thus, FIGS. **40** and **41** illustrate a new arrangement for the beams relative to those described above, but the beams include the same features for attachment, hanging, maintaining in folded configuration, and the like which have been discussed herein.

(107) Referring to FIGS. **44** and **45**, an architectural fixture **1100** is illustrated in accordance with an embodiment of the present invention. The architectural fixture **1100** formed by attaching several of the architectural fixtures **1000** of FIGS. **40** and **41** together. That is, the architectural fixtures **1000** are coupled together with joiner brackets **1130**, which have the same structure and function in the same way as the joiner brackets **130** described above. That is, the joiner brackets **1130** slidably engage the notches in adjacently positioned beams to connect the beams together.

(108) Referring to FIGS. **46-48**, an architectural fixture **1200** is illustrated in accordance with an embodiment of the present invention. The architectural fixture **1200** comprises a plurality of beams

1210, 1220, 1230, 1240, 1250, 1260 that are attached together to form a star or asterisk-like shape. While there are six of the beams **1210, 1220, 1230, 1240, 1250, 1260** illustrated in the exemplified embodiment, the invention is not to be so limited and more or less than six beams could be used in this arrangement in other embodiments utilizing the attachment features described below. Moreover, it should be appreciated that several of the architectural fixtures **1200** may be coupled together using methods and techniques described herein to form a large fixture with triangular shaped cells defined between the beams.

(109) The details of the architectural fixture **1200** will be described with reference to the beam **1210**, but it should be understood that the other beams **1220, 1230, 1240, 1250, 1260** have an identical structure to the beam **1210**. The beam **1210** extends along a longitudinal axis A-A. The beam **1210** a first sidewall **1211** that extends along the longitudinal axis A-A from a first end **1212** to a second end **1213** and a second sidewall **1214** that extends along the longitudinal axis A-A from a first end **1215** to a second end **1216**. Furthermore, the beam **1210** comprises a first upper flange **1217** that extends from the second end **1213** of the first sidewall **1211** to a distal end and a second upper flange **1218** that extends from the second end **1215** of the second sidewall **1214** to a distal end. The first and second upper flanges **1217, 1218** may be connected to the first and second sidewalls **1211, 1214**, respectively, by a living hinge in some embodiments, although this is not required in all embodiments. The first and second upper flanges **1217, 1218** extend from the first and second sidewalls **1211, 1214** towards one another in the exemplified embodiment.

(110) The architectural fixture **1200** comprises brackets **1270** that hold one of the upper flanges of one of the beams and one of the upper flanges of an adjacently positioned beam. For example, FIG. **46** illustrates the bracket **1270a** holding the second upper flange **1218** of the beam **1210** and one of the upper flanges of the beam **1220**.

(111) The beam **1210** (and the other beams as well) also comprises a first lower flange **1219** and a second lower flange **1280**. In the exemplified embodiment, a retention device **1290** is pulled through the lower flanges of all of the beams **1210-1260** and pulled tight to lock all of the beams together in the desired arrangement. The retention device **1290** may be a zip tie or other device capable of coupling the lower flanges **1219** of the respective beams **1210, 1220, 1230, 1240, 1250, 1260**.

(112) In the exemplified embodiment, the architectural fixture **1200** comprises a central hub **1295** with a plurality of notches **1296** (the number of notches should equal the number of beams, and thus in the exemplified embodiment there are six of the notches **1296**). One of the brackets **1270** is disposed within each of the notches **1296**. The central hub **1295** may be omitted in some embodiments.

(113) Referring now to FIGS. **49-51C**, an architectural fixture **1300** will be described in accordance with yet another embodiment of the present invention. The architectural fixture **1300** uses similar techniques to that described above with reference to FIGS. **46-48** for achieving the coupling of the various beams together, although the beams are in a different configuration. In particular, in this embodiment the architectural fixture **1300** comprises six beams **1310, 1320, 1330, 1340, 1350, 1360**. The first three beams **1310, 1320, 1330** are coupled together using brackets **1370** much like the brackets **1270** described above. Similarly, the second three beams **1340, 1350, 1360** are coupled together using brackets **1370** much like the brackets **1270** described above. Furthermore, the beam **1330** is coupled to the beam **1340** using joiner brackets **1390** which are identical to the joiner brackets **130** described above. That is, the joiner brackets **130** nest within notches in the inner surfaces of the beams **1330, 1340** adjacent to their ends to couple the beams **1330, 1340** together.

(114) Each of the beams **1310, 1320, 1330, 1340, 1350, 1360** extends along a longitudinal axis. By way of example, a longitudinal axis A-A is illustrated as extending along the beam **1310** and a longitudinal axis B-B is illustrated as extending along the beam **1330**. A first plane P1 lies along a bottom side of the beam **1310**. A second plane P2 lies along a bottom side of the beam **1330**. The beams **1310, 1320, 1330, 1350, 1360** are all angled with respect to each other. The beams **1330,**

1340 are aligned with each other. The planes formed by the bottom sides of the beams **1310**, **1320**, **1330**, **1350**, **1360** are non-parallel to one another.

(115) In this embodiment, the beams **1310-1360** have bottom flanges that are articulatable to allow the faces of different angles to come together in a clean way. The bottom flanges may be attached together with a retention device as described above. FIG. **51A** is a bottom view which illustrates the bottom flanges of the three beams **1340**, **1350**, **1360** interfacing in a clean manner. Furthermore, FIG. **51B** illustrates one of the beams **1310-1360** in an unfolded configuration. There are holes **1391** in the bottom flanges **1392** which allow for the retention device to pass through. Moreover, the remaining lines indicated on the unfolded beam are fold lines (which would be crease lines, pre-weakened lines formed by cutting or thinning of the material, or the like), which should indicate the manner in which the beam is manipulated to alter it from its unfolded configuration to its folded configuration.

(116) The architectural fixture **1300** and related embodiments allow a way to create a system of connected 3-way beams that can create an undulating three-dimensional surface out of a multitude of the 3-way beams. FIG. **51C** is one such example of an undulating three-dimensional architectural fixture formed using the techniques described with regard to this particular embodiment.

(117) FIG. **52A** illustrates the architectural fixture **100** of FIG. **1** with an indication that the architectural fixture **100** may have any length. FIGS. **52B**, **53**, and **54** illustrates the architectural fixture **100** of FIG. **5** with an indication that the architectural fixture **100** may have any length.

(118) The features described herein may be combined and mixed among the various embodiments. For example, although attachment brackets (such as **180A** and **180B** in FIGS. **1** and **5**) are not shown in all embodiments, they could be incorporated into any and all of the embodiments disclosed herein for purposes of hanging the architectural fixtures from a support surface. Thus, any of the features described with reference to one of the embodiments may be incorporated into any of the other embodiments to optimize use, function, aesthetic, or the like.

Claims

1. An architectural fixture assembly comprising: a beam extending along a longitudinal axis, the beam comprising: a bottom side; a first sidewall extending from the bottom side to a distal end and having an outer surface, an inner surface, and a first channel formed into the inner surface, the first channel comprising a first floor and a first roof, the inner surface of the first sidewall having a first portion above the first roof of the first channel and a second portion below the first floor of the first channel, the first channel being recessed relative to the first and second portions of the inner surface of the first sidewall; and a second sidewall extending from the bottom side and having an outer surface, an inner surface, and a second channel formed into the inner surface, the second channel comprising a second floor and a second roof, the inner surface of the second sidewall having a first portion above the second roof of the second channel and a second portion below the second floor of the second channel, the second channel being recessed relative to the first and second portions of the inner surface of the second sidewall; wherein the first and second sidewalls are configured to be folded relative to the bottom side between an unfolded configuration and a folded configuration; a first fastener located within the first channel; a second fastener located within the second channel; and a first securing element having a first securing feature that is configured to engage the first fastener within the first channel and a second securing feature that is configured to engage the second fastener within the second channel to maintain the beam in the folded configuration.

2. The architectural fixture assembly of claim 1 further comprising: the first roof and the first floor of the first channel extending from the inner surface of the first sidewall towards the outer surface of the first sidewall; the second roof and the second floor of the second channel extending from the

inner surface of the second sidewall towards the outer surface of the second sidewall; wherein the first fastener is oriented vertically between the first roof and the first floor of the first channel; and wherein the second fastener is oriented vertically between the second roof and the second floor of the second channel.

3. The architectural fixture assembly of claim 1 further comprising: the first sidewall having a first distal end, the first fastener extending through the first distal end of the first sidewall and into and through the first channel, wherein the first fastener is not visible along the outer surface of the first sidewall; and the second sidewall having a second distal end, the second fastener extending through the second distal end of the second sidewall and into and through the second channel, wherein the second fastener is not visible along the outer surface of the second sidewall.

4. The architectural fixture assembly of claim 1 wherein the first securing element is a flat plate comprising a planar top surface, a planar bottom surface that is parallel to the planar top surface, a first side edge extending between the planar top and bottom surfaces, and a second side edge extending between the planar top and bottom surfaces, the first and second side edges facing in opposite directions, the first securing feature comprising a first recess in the first side edge that is configured to receive the first fastener and the second securing feature comprising a second recess in the second side edge that is configured to receive the second fastener.

5. The architectural fixture assembly of claim 1 further comprising a first attachment bracket configured to engage the first channel and the second channel, the first attachment bracket being a distinct component from the first securing element, the first attachment bracket being configured to be mounted to a ceiling or a wall to hang the beam from the ceiling or the wall.

6. The architectural fixture assembly of claim 5 wherein the first attachment bracket comprises a central portion, a first arm portion extending from a first side of the central portion, and a second arm portion extending from a second side of the central portion, the first and second arm portions configured to nest within the first and second channels of the beam, respectively.

7. The architectural fixture assembly of claim 5 further comprising: the first securing element located adjacent to a first end of the beam and a second securing element located adjacent to a second end of the beam that is opposite the first end of the beam; and wherein the first attachment bracket is located axially in between the first and second securing elements.

8. The architectural fixture assembly of claim 6 further comprising the central portion of the first attachment bracket comprising an aperture that is configured to receive either a rod or a cable that is affixed to the ceiling to hang the beam from the ceiling or fasteners for mounting the first attachment bracket directly to the wall or the ceiling.

9. The architectural fixture assembly of claim 1 further comprising: the first sidewall extending from the bottom side to a first distal end, the first portion of the inner surface of the first sidewall extending from the first roof of the first channel to the first distal end of the first sidewall, and the second portion of the inner surface of the first sidewall extending from the first floor of the first channel to an upper surface of the bottom side; and the second sidewall extending from the bottom side to a second distal end, the first portion of the inner surface of the second sidewall extending from the second roof of the second channel to the second distal end of the second sidewall, and the second portion of the inner surface of the second sidewall extending from the second floor of the second channel to the upper surface of the bottom side.

10. The architectural fixture assembly of claim 1 wherein the bottom side and the first and second sidewalls of the beam define a cavity, and wherein the beam comprises a connection key at a first end of the beam, the connection key forming a boundary of the cavity along the longitudinal axis.

11. The architectural fixture assembly of claim 1 further comprising: a first notch in the inner surface of the first sidewall at a first end of the first sidewall; a second beam extending along a second longitudinal axis, the second beam comprising: a bottom side; a first sidewall extending from the bottom side; a second sidewall extending from the bottom side; and a second notch formed into the inner surface of the first sidewall of the second beam at a second end of the first

sidewall; a joiner bracket engaging the first notch of the beam and the second notch of the second beam, the joiner bracket securing the first end of the first sidewall of the beam to the second end of the first sidewall of the second beam.

12. The architectural fixture assembly of claim 1 further comprising: a second beam extending along a second longitudinal axis, the second beam comprising: a bottom side; a first sidewall extending from the bottom side; and a second sidewall that extends from the bottom side; and wherein the beam comprises a first monolithic, unitary sheet that comprises the bottom side, the first sidewall, and the second sidewall of the beam; and wherein the second beam comprises a second monolithic, unitary sheet that comprises the bottom side, the first sidewall, and the second sidewall of the second beam.

13. The architectural fixture assembly of claim 12 wherein the second beam is coupled to the beam such that the second longitudinal axis of the second beam is oriented at an angle to the longitudinal axis of the beam.

14. An architectural fixture assembly comprising: a first beam extending along a first longitudinal axis, the first beam comprising: a bottom side; a first sidewall extending from the bottom side, the first sidewall having an outer surface and an inner surface; a second sidewall extending from the bottom side; a first slot extending through the first sidewall from the outer surface to the inner surface; a second slot extending through the first sidewall from the outer surface to the inner surface; a third slot extending through the first sidewall from the outer surface to the inner surface, the third slot located between the first and second slots; a second beam and a third beam each comprising: a bottom side; a first sidewall extending from the bottom side; and a second sidewall extending from the bottom side; wherein a portion of the first sidewall of the second beam is received in the first slot of the first beam and a portion of the second sidewall of the second beam is received in the third slot of the first beam to couple the second beam to the first beam, the second beam comprising a second longitudinal axis that is oblique to the first longitudinal axis of the first beam; and wherein a portion of the first sidewall of the third beam is received in the second slot of the first beam and a portion of the second sidewall of the third beam is received in the third slot of the first beam to couple the third beam to the first beam, the third beam comprising a third longitudinal axis that is oblique to the first longitudinal axis of the first beam.

15. The architectural fixture assembly of claim 14 wherein the first slot extends through the first sidewall of the first beam at a first oblique angle relative to the longitudinal axis and the third slot extends through the first sidewall of the first beam at a second oblique angle relative to the first longitudinal axis, and wherein the third longitudinal axis is perpendicular to the second longitudinal axis.

16. The architectural fixture according to claim 1 further comprising: the first fastener extending along a first axis that is perpendicular to the bottom side of the beam, the first fastener not being visible along the outer surface of the first sidewall; and the second fastener extending along a second axis that is perpendicular to the bottom side of the beam, the second fastener not being visible along the outer surface of the second sidewall.

17. The architectural fixture according to claim 14 further comprising: the first slot having a first width, the second slot having a second width, and the third slot having a third width, the third width being greater than the first and second widths.

18. The architectural fixture according to claim 14 further comprising: the second longitudinal axis of the second beam being orthogonal to the third longitudinal axis of the third beam.

19. The architectural fixture according to claim 14 further comprising: the first beam defining a first cavity, the second beam defining a second cavity, and the third beam defining a third cavity; wherein end portions of each of the first and second sidewalls of each of the second and third beams are located within the first cavity of the first beam; wherein the end portion of the second sidewall of the second beam is located within the third cavity of the third beam; and wherein the end portion of the second sidewall of the third beam is located within the second cavity of the

second beam.

20. An architectural fixture assembly comprising: a first beam extending along a first longitudinal axis and comprising a bottom side and first and second sidewalls extending from the bottom side, the first sidewall comprising a first slot, a second slot, and a third slot located between the first and second slots; a second beam extending along a second longitudinal axis and comprising a bottom side and first and second sidewalls extending from the bottom side, the second beam coupled to the first beam via engagement between the second beam and the first and third slots of the first beam such that the second longitudinal axis is oblique to the first longitudinal axis; and a third beam extending along a third longitudinal axis and comprising a bottom side and first and second sidewalls extending from the bottom side, the third beam coupled to the first beam via engagement between the third beam and the second and third slots of the first beam such that the third longitudinal axis is oblique to the first longitudinal axis.
